

Constraints on modal flavor colexification

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Abstract

Modal items are known to often show ambiguity with respect to modal flavor. For example, English *can* can express both deontic and circumstantial flavor. To date, however, it remains unstudied whether there are particular sets of modal flavors that are more frequently colexified than others across the world’s languages. Here, we avail ourself of the recent database of Uegaki et al. (2024) to uncover natural classes of modal flavor colexification, and to evaluate the degree that these natural classes can be captured using different kinds of ontological structures.

Keywords: modal flavor, colexification, typology, semantic space, natural language ontology

1 Introduction

Modal items, used to express statements of possibility and necessity, can be characterized with respect to *force* and *flavor*. Force refers to the strength of quantification, depending on whether the modal makes a statement about what is possible (‘existential’ or ‘weak’ force) or what is necessary (‘universal’ or ‘strong’ force).¹ Flavor refers to the domain of quantification. Examples (1)–(4) provide English universal and existential modals exemplifying four commonly distinguished modal flavors: epistemic (relating to knowledge), deontic (relating to rules or laws), teleological (relating to goals), and circumstantial (relating to physical abilities).² Further flavors are sometimes posited, including bouletic (relating to desires).

Modal items are known to often show ambiguity with respect to modal flavor. For example, English *have to* is specified as having universal force, but can be used with any

¹ Additional force specifications, such as weak necessity, have been discussed in the literature, sometimes derived from strong modals via a further domain restriction (von Stechow and Iatridou, 2008).

² If it is difficult to conceptualize what a universal circumstantial modal would be, it can be useful to reason with de Morgan’s laws on existential circumstantial modals: *I can’t not laugh* = *I have to laugh*.

of the four flavors listed above. English *may* and *can* are specified as bearing existential force; *may* is ambiguous between epistemic and deontic flavor, while *can* is ambiguous between deontic, teleological, and circumstantial flavor.

- | | | |
|-----|---|----------------|
| (1) | a. He has to be the murderer! | EPISTEMIC |
| | b. I may have made a mistake... | |
| (2) | a. You have to clean your room! | DEONTIC |
| | b. You {may/can} see your friends. | |
| (3) | a. You have to take the bus to get to LGA. | TELEOLOGICAL |
| | b. You can take the A or E metro to get to JFK. | |
| (4) | a. I just have to laugh. | CIRCUMSTANTIAL |
| | b. Superman can lift a car. | |

Looking only at the English data, one may hypothesize that modals in general specify modal force, but may display ambiguities with respect to modal flavor. An influential result from the literature, however, shows that patterns of modality go beyond the flavor ambiguity of English-like systems. Notably, modals in a small number of other languages have been shown to also display ambiguity of force. In St’át’imcets (Lillooet Salish), for example, the modal *k’a* is specified for epistemic flavor while the separate modal clitic *ka* is specified for deontic flavor. Both modals are compatible with existential and universal force (Rullmann et al., 2008). Patterns such as these have lead researchers to theorize constraints on the typological profile of attested patterns, focusing notably on the (non-)interaction of force and flavor. Such constraints include the hypothesis of ambiguity along one axis at a time (Nauze, 2008), or the independence of force and flavor (Steinert-Threlkeld et al., 2023), which posits that a given modal item lexifies an exhaustive crossing of its force and flavor specifications.

Setting aside modal force, flavor alone already presents a complex landscape of colexification, which may itself be subject to systematic constraints and tendencies. Taking a sample of just eight modal verbs in English, for example, we observe that different modal items lump together modal flavors in different ways, as seen in (5), where we abbreviate the four flavors above with their initial. If we were to do the same exercise across many languages, what patterns would emerge?

(5) Colexification of modal flavor in English

- | | |
|------------------|----------------|
| a. must: ED | e. may: ED |
| b. have to: EDTC | f. can: DTC |
| c. need to: DTC | g. could: ET |
| d. should: DT | h. might: E(T) |

This is the topic that we seek to address in the present paper. Specifically, we ask two related questions: first, are there certain *natural classes* of modal flavors? Second, what does this tell us about the *ontological structure* of modal flavor? Specifically, if sets of modal flavors frequently pattern together, we take this as evidence that the flavors share some semantic feature. The nature of these features, and their relation to each other can thus in principle be backwards-engineered from typological data.

As our empirical base, we employ the recently created database of modal colexifications by Uegaki et al. (2024), related to a similar initiative by Guo et al. (2022). The Uegaki et al. database reports semantic judgments regarding the force-flavor combinations available to a couple hundred modal items across 24 languages. Due to the translation-based methodology on which it is based, the Uegaki et al. database has relatively sparse inclusion of negative judgments, meaning that a majority of the entries indicate a lack of direct evidence one way or the other. In a first step, we thus replicate data collection for three languages included in Uegaki et al. (2024) using a different methodology, in which acceptability judgments are exhaustively provided for each item-meaning pair. The results thus provide insight regarding the best way to treat the undetermined entries in the remainder of the Uegaki et al. database.

We then conduct several analyses to identify natural classes that emerge among modal flavors across the 24 languages in the database. We compare these natural classes to categories of modal flavor that have previously been proposed in the literature based on independent evidence, and we evaluate the degree to which these natural classes can be modeled using different kinds of ontological structures, including linear orderings, semantic maps, and feature geometries. In terms of natural classes, we provide new independent evidence for the epistemic-root divide that is commonly proposed in the literature, as well as for a class encompassing deontic and teleological modality, similar to the ‘priority’ modals of Portner (2009). We show that, contra what might be expected on a cartographic analysis of modality (Cinque, 1999), it is impossible to provide a linear order of modal flavors that captures the observed patterns of colexification. Similarly, we show that semantic maps based on pairwise relations like those of Haspelmath (1997) or van der Auwera and Plungian (1998) similarly lack the necessary power to capture the descriptive generalizations. These findings support an analysis in terms of semantic features, possibly organized into a feature geometry. We discuss the implications of these findings, as well as the inherent limitations of our analysis due to the nature of our dataset.

2 Colexification

Following conventions in the literature, we use the term *colexification* to describe a situation in which different meanings are expressed by the same word. The term is deliberately a catch-all term encompassing ambiguity, polysemy, and underspecification. Work on the lexical semantics of both content words (Gärdenfors, 2004, 2014) and function words (e.g. Chemla et al., 2019; Haspelmath, 2023) has demonstrated that colexification is subject to principled constraints.

One such constraint is that of *connectedness*: if an exponent ϕ (typically taken to be a monomorphemic word) applies to concepts A and C , and concept B is defined as being situated ‘between’ A and C , then ϕ applies to B . For example, in the color domain, if the same color term applies to things that are blue and things that are green, then it must also apply to things that are blue-green.

A second factor influencing colexification is that of conceptual *relatedness*. Concepts that are closer in a semantic space—e.g. FINGER/TOE compared to FINGER/HEAD—have been observed to be more commonly colexified (Xu et al., 2020), although only to the extent that this colexification preserves contextually-relevant semantic distinctions (Brochhagen and Boleda, 2022): e.g. the concepts RIGHT and LEFT are unlikely to be

colexified, despite having very high conceptual relatedness.

The color space is a clear and well-studied domain in which a single underlying representation can give rise to systematic typological variation. Notably, in this domain, non-linguistic work on human perception has motivated a irregularly-shaped space of colors; Figure 1a depicts the CIELAB color space, in which Euclidean distance corresponds roughly to perceptual distance. Linguistic work has shown that the world’s languages show implicational generalizations regarding the way in which colors are labeled (Berlin and Kay, 1969). For example, some languages have more color words than others, but it has been found that if a given language has a word for the concept BLUE, then it generally also has a word for the concept RED. In recent work, it has been shown that these typological generalizations can be derived by principles of efficient communication operating on the space of color perception (Regier et al., 2007; Zaslavsky et al., 2018).

Color is thus a domain in which the underlying semantic space is assumed to be continuous (and Euclidean). But connectedness and related principles can apply equally well to discrete domains with complex structure. The closest discrete correlate to the continuous example of color is probably the tradition of modeling meaning using *semantic maps*, in which a semantic space is organized as a discrete graph representing pairwise relations between concepts (Haspelmath, 2003). As an illustration, Figure 1b provides an example from Haspelmath (1997) intended to capture how the space of indefinite meanings is structured. Critically, this representational convention is predicated on the same representational constraint of connectedness. Just as connectedness applies to concepts in the continuous color space, it applies to the discrete space of indefinites: if a word applies to a SPECIFIC KNOWN referent and an IRREALIS NON-SPECIFIC referent, then it should also apply to a SPECIFIC UNKNOWN referent. Likewise, calculations of communicative efficiency are well-defined and can be applied to discrete conceptual domains (Imel and Steinert-Threlkeld, 2022).

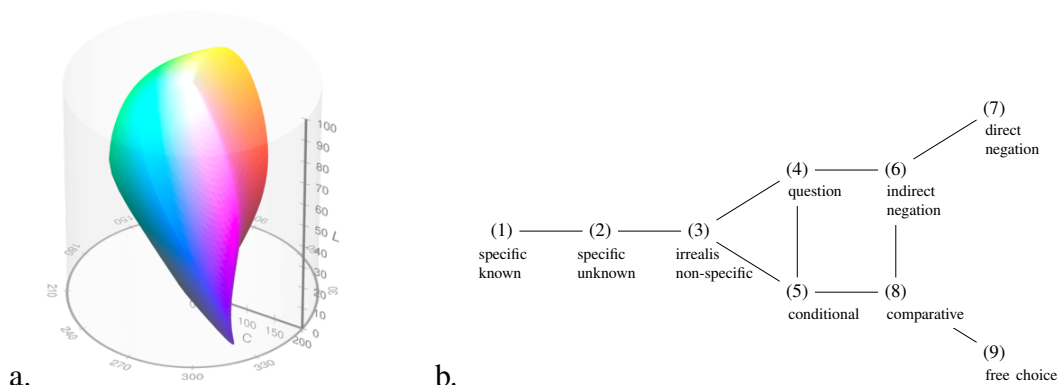


Figure 1: Examples of continuous and discrete meaning spaces. Figure 1a of CIELCHab color space is by Michael Horvath and Christoph Lipka under CC-BY-SA-4.0.

As seen above, one methodological approach used to examine lexification constraints consists of assuming a particular ontological structure, and using colexification to explore the principles of communicative efficiency that shape lexical inventories. This, for example, is the approach taken by Zaslavsky et al. (2018), who assume a semantic space coming from work on human color perception. For modal meaning, it is also the approach of Imel and Steinert-Threlkeld (2022), who assume a semantic space with two dimensions

(force $\times$ flavor) and nominal (i.e., unstructured) levels.

An alternative approach consists of doing the opposite: to assume particular constraints on colexification, and explore the possible ontological structures that lead to a particular typology. This, for example, is explicitly the approach of Haspelmath (1997, 2003) who assumes that meanings are connected and explores semantic maps that adhere to the typological data. In the present work, we apply the latter approach to modal items, focusing specifically on ambiguity of modal flavor. In addition to semantic maps, we consider alternative ontological structures for discrete meaning spaces, including total orderings and feature geometries, discussed further in Section 3.3. We consider a range of possible colexification constraints, including connectedness, and evaluate how well the various ontological structures capture the patterns of colexification that appear in the cross-linguistic data.

3 The ontology of modal flavor

Although the term colexification collapses ambiguity, polysemy, and underspecification, a good case can be made that—in English at least—modal colexification is a case of ambiguity. Specifically, as observed by Bassi and Jeretic (2023), the flavor of a modal item generally cannot shift under ellipsis. In (6), for example, both epistemic and deontic interpretations are possible for the verb *may*, but the same interpretation must be preserved in both clauses:

- (6) John may laugh, but not Bill.
Not possible: John might laugh; Bill is not allowed to.

This ambiguity could arise from one of several sources. One possibility is lexical ambiguity: there are simply two verbs, *may*₁ and *may*₂, and each specifies a different flavor. A second possibility, essentially the classical proposal of Kratzer (1977, 1981), is that these are cases of pronominal ambiguity. A simple such definition of *may* is provided in (7). In this definition, DOM is a free pronoun over sets of worlds. On the full analysis of Kratzer, this set of worlds is derived in two steps: first a contextually-supplied *modal base* provides a set of accessible worlds; second, a contextually-supplied *ordering source* provides a means to rank these accessible worlds. The domain DOM is the set of ‘best’ worlds among the accessible worlds with respect to the ordering source.

$$(7) \quad \llbracket \text{may} \rrbracket^w = \lambda p_{\langle s, t \rangle} [\exists w' [w' \in \text{DOM}(w) \wedge p(w')]]$$

On such a framework, the set of flavors available to a given modal item can be viewed as a presupposition on the values available to its contextually-supplied arguments. Just as individual pronouns may have presuppositions that restrict their possible values (e.g. English *she* can only refer to female individuals), a given modal item may restrict the possible values of DOM—or, in a more articulated view, the values of the modal base and the ordering source. Modeled this way, there are infinitely many values of DOM³, but this

³Indeed, if we allow DOM to be any set of possible worlds then, not only does it have infinitely many possible values, but, by Cantor’s theorem, the set of possible values for DOM is of a greater (infinite) cardinality than the set of possible worlds.

does not mean that there are infinitely many modal flavors. While context can in principle select any salient value as a domain of quantification, the presuppositions on modal items inject ontological constraints that categorize and structure this semantic space. For example, depending on context, one could evaluate a modal sentence with respect to the laws of France or with respect to the laws of Germany. These thus correspond to two distinct values of DOM, but the two (deontic) domains are never distinguished with different lexical items. On this view, modal flavors are not possible values of DOM, but rather categories that identify a semantic property shared across a range of modal domains. From this perspective, we can think of our case of colexification as involving the natural classes that may be covered by this presupposition.

3.1 Epistemic and root modality

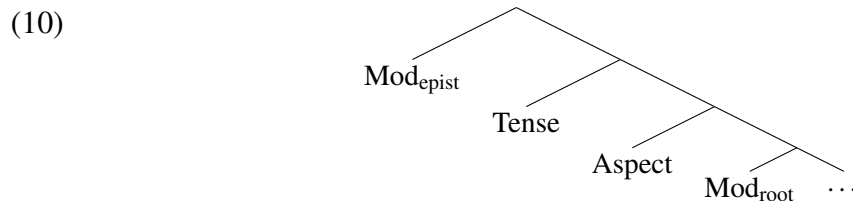
One well-known proposed natural class of flavor is category of *root* modality, encompassing deontic, teleological, and circumstantial flavors, to the exclusion of epistemic modality (Hacquard, 2006). The epistemic-root opposition has been motivated on both conceptual and empirical grounds.

First, it has been observed that epistemic flavor is speaker-oriented, while root modality is oriented to the subject or another participant of the asserted event, as shown in (8) (Zagona, 2008).

- | | | |
|-----|------------------------|-------------------------|
| (8) | a. It must be raining. | = knowledge of speaker |
| | b. John has to leave. | = obligation of subject |
| | c. John can juggle. | = ability of subject |

Second, an empirical basis for the epistemic-root opposition is found in the interaction of modality with tense, aspect, and negation. Epistemic modals generally take wide scope, while root modals tend to take narrow scope (Stowell, 2004). The interaction with tense is illustrated in (9). The sentence in (9a) necessarily relates to the knowledge of the speaker at the time of the utterance, while the one in (9b) necessarily relates to the circumstances in the world at the time of the trip to Paris. One influential explanation of these scopal facts has been offered in the form of a cartographic analysis (Cinque, 1999). This type of analysis posits that epistemic modals appear above Tense and Aspect in the syntactic structure, while root modals must merge below them, as in (10).

- | | | |
|-----|---|-------------------------|
| (9) | a. John had to be home at the time of the crime
(*but now it turns out he wasn't). | $\square > \text{Past}$ |
| | b. Mary had to take the train to go to Paris
(but now flights have been added). | $\text{Past} > \square$ |



Related evidence comes from the availability (or not) of stacked modals, and the interpretations that they can receive (Werkmann Horvat, 2021). For example, we have

seen that the English modals *may* and *have to* both allow epistemic and deontic readings; however, the combination of the two only allows the E>D reading, as shown in (11).

- (11) You may have to leave. E>D, *D>E
- a. The speaker entertains the possibility that the addressee is obliged to leave.
 - b. * The addressee is allowed to have the property that the speaker is certain that they are leaving.

The epistemic-root opposition has thus primarily been motivated on the basis of scopal properties exhibited by modal items. In what follows, we will investigate whether colexification patterns similarly reflect these two categories, and, if so, whether the same mechanisms can explain both the colexification facts and the scopal properties.

3.2 Further natural classes (beyond the epistemic-root divide)

A number of researchers have proposed further refinements of this typology of modal flavor either by proposing a tripartite distinction (instead of the bipartite epistemic-root divide) (Brennan, 1993; Hacquard, 2006; Portner, 2007, 2009), or by defining subclasses of root or epistemic modality.

Notably, Portner (2007, 2009) argues that what is traditionally viewed as a class of root modals should instead be viewed as (a) a class of *priority modals*, containing deontic, teleological, and bouletic flavors and (b) a class of *dynamic modals*, containing circumstantial flavor (among others). Intuitively, the class of priority modals is characterized by the fact that there is a relatively easy-to-identify ordering source that allows one to rank some worlds as better or higher priority than others. Empirically, Portner (2007) motivates this class based on parallels with imperatives. Imperatives have been shown to allow a variety of different functions, including orders and invitations, among others, as in (12)–(13). The fact that the same functions are available for priority modals (but not for other classes of modal items), motivates the proposal that they are built from the same semantic building blocks (for Portner, the ordering source of priority modals are To-Do Lists).

- (12) a. Get out of my office! (Order)
b. You should get out of my office.
- (13) a. Have a piece of fruit! (Invitation)
b. You should have a piece of fruit.

Portner also explicitly argues *against* the existence of a designated class of root modals, proposing instead that priority and dynamic modals are only epiphenomenally unified by their property of not being epistemic. This being said, it is worth noting that Portner (2009) does adopt a Kratzerian account on which epistemic and non-epistemic modals are constructed using different types of modal bases: epistemic bases for the former, and circumstantial bases for all non-epistemic flavors. Thus, even on Portner (2009)'s analysis, root modals are minimally definable by the fact that they exclusively share a particular primitive semantic feature, namely the circumstantial base.

A related subcategorization of root modality is the opposition of *participant-internal* and *participant-external* modality proposed by van der Auwera and Plungian (1998),

building on diachronic data from Bybee (2003). Intuitively, participant-internal modality is defined by circumstances internal to the individual (capacities, needs, and desires); participant-external modality is defined by circumstances external to the participant (laws, facts about the world). Like Portner (2009), this categorization creates a subcategory that includes deontic and teleological modality; it differs, however, in the placement of bouletic flavor within this typology. Portner (2009) classifies bouletic flavor as a priority modal, but it is defined based on participant-internal circumstances (desires); van der Auwera and Plungian (1998) might thus predict it to form a natural class with circumstantial modality (relating to capacities).

Characterization of modal flavors beyond the epistemic-root divide has also been conducted within the cartographic framework. Within this literature, modal flavors are not typically grouped into natural classes, but instead are taken as primitives that stand in a particular order on the syntactic spine, motivated by data such as ordering restrictions on modal items in various languages. Two such orders that have been proposed are provided in (14). While the inventory of flavors and their exact order are slightly different between the two, we note that both proposals share the property of ordering priority modals above other root modals.

- (14) a. EPISTEMIC > BOULETIC/VOLITIONAL > DEONTIC
 > DESIGN > CIRCUMSTANTIAL > ROOT (Bross and Hole, 2017)
 b. EPISTEMIC > PRIORITY (DEONTIC, BOULETIC, TELEOLOGICAL)
 > PURE POSSIBILITY > ABILITY > DISPOSITION (Werkmann Horvat, 2021)

There are further distinctions that fall outside the scope of the analyses of the present paper, due to the data that we have available; these are nevertheless mentioned here and discussed further in Section 6.5. First, the category of epistemic modality may have further sub-flavors; in particular, it has been proposed that epistemic modality may be further categorized as *subjective* or *objective* (Nuyts, 2001): subjective epistemic flavors are based on beliefs or knowledge specific to the speaker; objective epistemic flavors are based on more generally-accessible information. For example, Lyons (1977) observes that sentence (15) has two modal items, *certainly* and *may*, both of which appear to have epistemic flavor (despite having different force). Lyons (1977) proposes that the first modal expresses subjective epistemic modality and the second, objective epistemic modality. Some languages may have modal items that specifically identify these flavors.

- (15) Certainly he may have forgotten.

Second, it has been argued that uses of deontic modals can be classified as *ought-to-do* or *ought-to-be* (Brennan, 1993). In the former category, the modal places an obligation on (or gives permission to) the subject of the sentence, as in (16a); in the latter, the modal obliges a particular state to hold; the bearer of this obligation may be the speaker or addressee, as in (16b). Notably, *ought-to-be* uses of deontic modals share the property with epistemic modals that they relate to participants of the speech act.

- (16) a. John needs to clean his room.
 b. The speech needs to be ready for tomorrow.

These alternative proposals to a bipartite epistemic-root structuring of the flavor space therefore equip us with the following additional flavor oppositions: dynamic *vs* priority,

participant-internal vs participant-external, and subjective vs objective. We note, however, that the motivation for these categories is somewhat inconsistent, often relying more on conceptual grounds than on morphosyntactic reflexes. Considering colexification data will thus provide an independent source of evidence on which the cognitive reality of these categories can be evaluated.

3.3 Semantic primitives and their structures

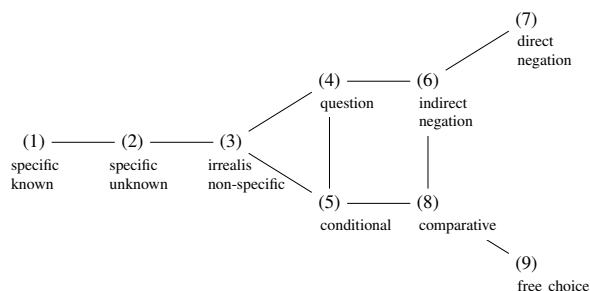
In other linguistic domains, a variety of different ontological structures have been proposed to capture natural classes of meanings. Each of these can be considered and evaluated in the context of modal flavor.

First, we have seen that cartographic approaches to modality posit that modal flavors are distributed along the syntactic spine (Cinque, 1999; Bross and Hole, 2017; Werkmann Horvat, 2021). On such an analysis, modal flavors can be attributed a *linear, one-dimensional ordering*, as in (17). Flavors may pattern together, forming a natural class, if they appear in a particular part of the syntactic spine; for example, for Cinque (1999) the category of root modals are those that appear below Tense and Aspect.

(17) Linear: $A > B > C > D$

A second possibility discussed above is the concept of the *semantic map*, in which semantic relatedness is represented via a graph of pairwise connections, again illustrated in (18) with the case of indefinites (Haspelmath, 1997). Meanings may pattern together, forming a natural class, if they appear in a connected subpart of the graph. We note that a semantic map is strictly less constrained than a linear order: a total ordering can be rewritten as a simple, linear semantic map, but not vice versa.

(18) Semantic map:

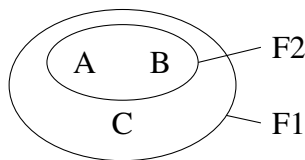


(Haspelmath, 1997)

A third alternative, foreshadowed above, is the use of semantic *features* to capture natural classes of meanings. Instead of treating meanings as atomic primitives, this view posits that meanings correspond to particular bundles of semantic features. This perspective allows one to model sets of natural classes that are impossible to model using semantic maps. In particular, because semantic maps are defined in terms of pairwise relations between meanings, it is impossible to define a natural class containing three or more elements without also positing subclasses containing pairs of elements. To see this, consider the example in (19). Here, Feature 1 defines a natural class containing meanings A, B, and C; Feature 2 defines a natural class containing meanings A and B. Critically, no

feature defines a natural class containing just A and C or just B and C. This set of natural classes cannot be modeled using a semantic map, since there is no way to connect C to A and B without being connected to either the one or the other.

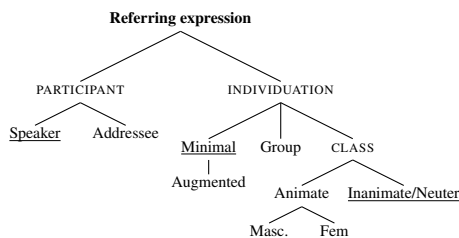
(19) Features:



There exists work that reanalyzes semantic maps in terms of the semantic features shared between different nodes in the graph. For example, Aloni and Degano (2023) analyze a subpart of the map of indefinite meanings in (18) as emerging from the constancy or variation of the indefinite’s value with respect to different variables. Different specification of these semantic parameters generate exactly the six logical possibilities for selecting a connected subpart of the leftmost three nodes in the map in (18). On such an analysis, the semantic map is not itself the underlying representation, but rather a structure emerging from the way different semantic features interact.

Additional semantic or morphosyntactic properties may place further constraints on the ways that features may interact. For example, it may be the case that one bundle of features is uninterpretable due to the presence of contradictory semantic constraints. Morphosyntactic constraints are perhaps best instantiated in the form of *feature geometry*, which represents implicational relations between features: if feature Y is a daughter of feature X, then an element can only bear feature Y if it also bears feature X. This is illustrated in (20) with an analysis of pronoun features (Harley and Ritter, 2002), where we observe, for example, that the feature CLASS (specifying gender) can only appear on referring expression that also bear the feature INDIVIDUATION (specifying number), reflecting the typological generalization that languages only show gender agreement if they also show number agreement.

(20) Feature geometry:



(Harley and Ritter, 2002)

For each of the ontological structures above, we will evaluate their ability to explain the patterns of modal flavor colexification that appear in the data set.

3.4 The independence of modal flavor?

One of the premises grounding the present paper is the assumption that modal flavor can, to a meaningful extent, be modeled as an independent semantic dimension with its own ontological structure.

Most saliently, our approach adheres to Steinert-Threlkeld et al. (2023)’s hypothesized Independence of Force and Flavor (IFF) universal, stated in (21). This constraint allows for modal items to be underspecified for force, for flavor, or both force and flavor, but it rules out interactions between the two dimensions—for example, a modal item whose only two uses are as a universal epistemic and an existential deontic. This assumption appears to be supported by cross-linguistic data and has been motivated based on communicative efficiency (Imel and Steinert-Threlkeld, 2022).⁴

(21) *The Independence of Force and Flavor Universal*

If a modal can express the pairs (force₁, flavor₁) and (force₂, flavor₂), then it can also express (force₁, flavor₂) and (force₂, flavor₁)

A second semantic factor that will be relevant to our analytic choices below regards the interaction of modal flavor with the polarity of the semantic environment in which an item appears. Notably, for English at least, it turns out that the flavors available to a given modal item at least occasionally depend on whether they appear in a positive or negative form. For example, the verb *can* generally does not allow an epistemic reading, but the negated form *can’t* does allow it, as shown in (22). Conversely, the verb *must* and adverb *necessarily* allow an epistemic reading in the positive form, the negative forms *mustn’t* and *unnecessarily* lose the epistemic interpretation, as shown in (23)-(24).

- | | | |
|------|---------------------------------------|------------|
| (22) | a. It can be raining. | *epistemic |
| | b. It can’t be raining. | ✓epistemic |
| (23) | a. John must be the culprit. | ✓epistemic |
| | b. John mustn’t be the culprit. | *epistemic |
| (24) | a. John is necessarily the culprit. | ✓epistemic |
| | b. John is unnecessarily the culprit. | *epistemic |

As far as we are aware, it remains unstudied whether these English examples are a relative anomaly, or whether the shifting of modal flavor in negative forms is a more general phenomenon across languages. To avoid possible confounds from polarity, we will thus restrict the scope of our main analysis in Sections 5.1-5.2 to positive uses of modal items. We return to the question of flavor-polarity interactions more explicitly in Section 5.3.

4 The data set

4.1 A database of force-flavor colexifications

To address the questions above, we employ the recently created database of modal colexifications reported in Uegaki et al. (2024), which provides semantic judgments regarding

⁴This assumption is not shared by all analyses of modals; a notable exception is Cinque (1999, p. 81) who posits four independent syntactic projections for (alethic) necessity, alethic possibility, deontic obligation, and deontic permission.

the force-flavor combinations available to a couple hundred modal items across 24 languages.⁵ Most judgments reported in Uegaki et al. (2024) have in fact been incorporated into the larger database of Guo et al. (2022), a related endeavor that similarly aims to collect force-flavor colexifications across many different languages. Guo et al. (2022) do not themselves provide new original data; rather, their database collects semantic judgments that have been previously reported in the literature on modality into a consistently coded format. The April 2023 release of the database thus included data from 40 languages total, 22 coming from Uegaki et al. (2024). In the present paper, we focus specifically on the data collected and reported by Uegaki et al. (2024). This decision is grounded on the assumption that results coming from a single research team will have less methodological variation, and on the fact that the choice of languages in Uegaki et al. (2024) was designed to sample a wide range of language families. In Section 5.2, we nevertheless replicate our analyses using the larger dataset of Guo et al. (2022).

The Uegaki et al. database was created using a translation-in-context task with native speakers of each of the 24 languages. Speakers were asked to provide a translation of a target sentence (containing a modal item), that would be natural and felicitous in a provided context, which identified a particular force and flavor combination. Speakers were invited to provide multiple translations, using different modal items and comment on the differences between these translations. Occasional acceptability judgments were asked for specific items in specific contexts to collect negative judgments. For each modal item and each force-flavor combination, the database thus includes a ‘1’ if the item can express that particular force-flavor combination, a ‘0’ if it cannot, and a ‘?’ if no direct evidence was collected to conclude one way or the other about whether the form can express the force-flavor combination. Table 1 provides sample entries from the Uegaki et al. database.

language	expression	force	flavor	can_express	polarity	full_form
dutch	zal	strong	epistemic	1	positive	zal
dutch	zal	weak necessity	epistemic	1	positive	zal
dutch	zal	weak	epistemic	?	positive	zal
dutch	zal	strong	deontic	?	positive	zal
dutch	zal	weak necessity	deontic	?	positive	zal
dutch	zal	weak	deontic	?	positive	zal
dutch	zal	strong	teleological	0	positive	zal
dutch	zal	weak necessity	teleological	?	positive	zal
dutch	zal	weak	teleological	?	positive	zal
dutch	zal	strong	circumstantial	?	positive	zal
dutch	zal	weak	circumstantial	?	positive	zal
dutch	zal	strong	bouletic	?	positive	zal
⋮						

Table 1: Sample entries in the Uegaki et al. (2024) database

Uegaki et al. (2024) recognize the importance of negative judgments, and accordingly do collect negative judgments for certain key pieces of data. Nevertheless, as suggested

⁵Languages include: Akan (ISO: aka), Basque (ISO: eus), Cantonese (ISO: yue), Dutch (ISO: nld), Farsi (ISO: pes), Greek (ISO: ell), Hausa (ISO: hau), Hebrew (ISO: heb), Hindi (ISO: hin), Hungarian (ISO: hun), Igbo (ISO: ibo), Japanese (ISO: jpn), Khmer (ISO: khm), Kîtharaka (ISO: thk), Korean (ISO: kor), Mandarin (ISO: cmn), Mapudungun (ISO: arn), Russian (ISO: rus), Spanish (ISO: spa), Tagalog (ISO: tgl), Telugu (ISO: tel), Thai (ISO: tha), Turkish (ISO: tur), Vietnamese (ISO: vie).

by Table 1, ?s still comprise the vast majority of the entries in the database: of the nearly 4000 entries, 18% are 1, 2% are 0, and a full 80% are ?s. Critically, if 80% of the dataset is indeed entirely undetermined, then it is likely very difficult to draw any meaningful generalizations about large-scale typological patterns.

Nevertheless, there are conceptual reasons to believe, in the context of the reported methodology, that force-flavor combinations marked ‘?’ are in fact not fully felicitous for a given modal item. In particular, when a subject is asked what item or items can appear in a particular context, the items that spring to mind are likely to be the ‘best’ items for that context. Even without direct negative evidence, items that are not mentioned are likely to not be *as* felicitous as those that are offered spontaneously. This effect is likely further amplified in a task in which judgments are repeated in a variety of different contexts, thus priming a list of items that subjects may consider using in the translation task.

In order to assess this hypothesis, we replicated Uegaki et al.’s study for three of their 22 languages, adopting a methodological paradigm in which subjects provided explicit acceptability judgments for each item-meaning pairing. A comparison of the findings from the two different methodologies is then used to decide how to treat the data from the remaining 19 languages, and more generally, to evaluate the degree to which lack of spontaneous production can and should be treated as *indirect* evidence for unacceptability in a particular methodological paradigm.

4.2 Replication with acceptability judgments

In order to better understand how to interpret this underdetermined data, we collected comprehensive acceptability judgments of item-meaning combinations for three of the languages included in the Uegaki et al. database: Hebrew, Mandarin, and Russian.⁶

For each force-flavor combination reported in Uegaki et al. (2024), we selected two corresponding contexts from Vander Klok’s (2021) modal verb questionnaire. These contexts distinguish between the three standard forces (necessity, weak necessity, possibility) and five flavors (epistemic, deontic, teleological, circumstantial, bouletic), that also appear in Uegaki et al. (2024). The contexts in Vander Klok (2021) are designed for fieldwork with Javanese speakers, so they often make reference to cultural aspects of Indonesian society. We thus modified the contexts to be easily understandable in a Western European cultural context (these are available in the Supplemental Materials). All consulted native speakers self-report to have been sufficiently familiarized with this cultural setting to evaluate the semantic conditions elicited by the context scenarios.

For each of the three languages, we conducted elicitation with two native speakers. All consultants were self-reported English–target language bilinguals, and elicitation was conducted in English. The English elicitation contexts were modified to include gaps (indicated as ‘???’ for the written survey) in place of the targeted modal item. Interviews were either conducted in-person or via online videoconferences. In-person elicitation

⁶Due to an oversight, judgments for deontic possibility were not collected for Hebrew or Mandarin. The first Russian speaker also objected to the decision in Uegaki et al. to treat *tzrikh*, *tzrikha*, and *tzarikh* as three separate items; consequently, only a single item was judged by both speakers, and the items were simply left out of the replication comparison. All other force-flavor combinations included in the Uegaki et al. database were otherwise tested for each item.

consisted of the interviewer reading out the elicitation contexts to the consultant in randomised order. Consultants were instructed to give judgments spontaneously but were not subjected to time limits.

Consultants were prompted to translate the English context into the target language, and to identify the item which they judged to correspond most closely to the gap ('???') included in the English context. Target language items were elicited in a two-step process: consultants were first asked to spontaneously provide their preferred item(s) to match the modal item gap, thus paralleling the methodology in Uegaki et al. (2024). In a second step, consultants were presented with all modal items included for positive contexts in the target language by Uegaki et al. (2024). Consultants were then asked to rate the acceptability of each of these items for every context. Consultants assigned '1' to all items judged as acceptable, '0' to unacceptable items, and an intermediate judgment (coded here as '0.5') to items which they judged to be marginally acceptable or only acceptable in a different flavor context than the one targeted by the elicitation context at hand. All productions from Step 1 and judgments from Step 2 are available as Supplemental Materials.

Although the two contexts selected for each force-flavor combination were intended to target the same modal meaning, subjects did not always assign the same acceptability judgment to a given modal item in each of the two contexts. Moreover, the two speakers of each language did not always agree with respect to the acceptability judgment assigned for each modal item. In light of these within- and between-subject disagreements, we considered that a particular modal item is able to express a particular meaning if at least one speaker accepts the meaning for at least one context. Concretely, we select the maximum rating (0, 0.5, or 1) among the four responses provided for a given item with a given meaning. This analytical choice has the risk of overestimating the meanings that are available to the target modal items, but this is the more conservative direction to err when evaluating the hypothesis that items marked ? in the Uegaki et al. dataset are generally not felicitous.

Tables 2–4 present the results of the methodological comparison. No entries in the Uegaki et al. database for Mandarin were marked as '0', so these lines do not appear in Table 3. Encouragingly, comparison of methodologies yields a rather similar pattern across all three languages. In all three languages, the large majority of entries that appear as ?s in the Uegaki et al. database are judged as unacceptable (0) or at best borderline (0.5); specifically, this accounts for 84% (Russian), 87% (Hebrew), or 93% (Mandarin) of '?' entries in the original database. Moreover, across all three languages, the number of '?' entries that were judged as available in our replication task was of a similar magnitude to the number of '1' entries that were judged as unacceptable or borderline in the replication task. In other words, the amount of noise arising from inter-subject variability appears to be greater than any methodological bias arising from misclassifying ?s as unacceptable when they are in fact available. For present purposes, we thus conclude that it is methodologically sound to treat ?s as 0s in the Uegaki et al. database, and to let any noise average out with large enough data sets. This is thus the assumption that we adopt in our subsequent analyses: we consider that a modal item is able to express only those meanings that are marked with a '1' in the Uegaki et al. database.

Finally, as noted above, our replication task uncovered a certain degree of intra-speaker variability, on which two contexts, intended to identify the same force-flavor combination, were nevertheless assigned different acceptability judgments by the same

speaker for the same item. This observation is compatible with the possibility that the three forces and five flavors that are employed by Vander Klok (2021) and Uegaki et al. (2024) might not be sufficient to describe the meaning of all the modal items studied, and there may be a need to create further subcategories of modal meaning. For present purposes, we will nevertheless continue to assume an ontology with only five modal flavors, as this assumption is built into the database that we have to work with, but the assumption will be discussed further in Section 6.5.

Uegaki et al.	Replication	Count	Percent	
1	1	12	86%	86%
1	0.5	1	7%	14%
1	0	1	7%	
?	1	12	13%	87%
?	0.5	15	16%	
?	0	66	71%	
0	1	0	0%	100%
0	0.5	1	33%	
0	1	2	67%	

Table 2: Replication of Hebrew data from the Uegaki et al. database.

Uegaki et al.	Replication	Count	Percent	
1	1	22	81%	81%
1	0.5	4	15%	19%
1	0	1	4%	
?	1	7	7%	93%
?	0.5	16	17%	
?	0	71	76%	

Table 3: Replication of Mandarin data from the Uegaki et al. database.

Uegaki et al.	Replication	Count	Percent	
1	1	19	76%	76%
1	0.5	2	8%	24%
1	0	4	16%	
?	1	15	16%	84%
?	0.5	7	7%	
?	0	72	77%	
0	1	1	100%	100%
0	0.5	0	0%	
0	0	0	0%	

Table 4: Replication of Russian data from the Uegaki et al. database.

4.3 Further methodological choice points

Two further methodological choices and their motivation should be mentioned. First, the Uegaki et al. (2024) database is notable in that it systematically codes for *polarity*, indicating which modal forces and flavors are available for a given item in positive environments and in negative environments. As we saw in Section 3.4, the polarity of the environment in which an modal item appears may in fact influence the flavors available to it; for example, English *can't* allows an epistemic reading but *can* does not. With respect to colexification, the two instances (*can* and *can't*) could in principle be considered to be either a single item, or two separate items. It is also perfectly possible that the optimal choice between these two options is different depending on the language and item in question. Thus, to avoid over-ascription of ambiguity and to avoid double counting, we made the decision to analyze only modal items appearing in positive environments in our main analyses, presented in Sections 5.1-5.2. We return to the interaction of negation and modal flavor more explicitly in Section 5.3.

Second, the Uegaki et al. (2024) database codes lexical roots, but also other morphology, such as aspect. Different aspectual inflections may influence the flavors available to a modal item (Matthewson and Truckenbrodt, 2018). For example, the Hausa modal *dole* is listed as allowing epistemic flavor when inflected with imperfective or perfective aspect, and as allowing deontic, teleological, circumstantial, and bouletic flavor when inflected with the subjunctive. As above, one must decide whether to count *dole*+IPFV and *dole*+SUBJV as the same item or as different items. Here, our decision was to count both inflections as instances as the same item, and merge morphological variants when listed as the same item in the original Uegaki et al. (2024) database.

Finally, bouletic modality appeared significantly less frequently than other flavors (epistemic, deontic, teleological, and circumstantial), appearing as a possible meaning for only 43 items, including 12 items that *only* express bouletic modality and 14 that can express *any* modal flavor (EDTCB). This leaves only 17 items with non-trivial flavor profiles that can be meaningfully interpreted as bearing on the natural classes of modal flavor (since mono-lexifications and total lexifications are natural classes on any theory). In order to avoid spurious results arising from sparseness of data, we thus decided to leave bouletic modality out of quantitative analyses below. Concretely, we categorized modal items as though bouletic flavor was not included in the dataset; an item with, for example, the flavor profile DTCB would thus be counted as an instance of the flavor profile DTC. Items that only express bouletic modality were left out of the analyses below. We revisit this analytic decision in 6.3, which discusses the (limited) information that is available when bouletic modality is included in the analyses.

These decisions are not trivial, and there may be reasonable theoretical arguments to make different decisions than the ones that we made here. As reported below, however, it turns out that the trends in the colexification data are relatively robust, and that making different choices on these methodological choice-points does not in fact have a notable effect on the main results presented below.

5 Analyses

5.1 Gap analysis

As described in Section 3.3, one possible ontological structure for modal flavors is a *one-dimensional ordering*. As discussed above, such a structure is necessarily assumed by cartographic approaches on which modal flavors are distributed across the syntactic spine (Cinque, 1999; Bross and Hole, 2017; Werkmann Horvat, 2021). Furthermore, as the most restricted ontological structure, it is also a reasonable null hypothesis. In our first analysis, we ask whether there is a one-dimensional ordering of flavors that clearly works best to explain the colexification data.

In particular, we adopt the standard assumption that colexifications should be *connected* (Gärdenfors, 2004; Chemla et al., 2019). For example, we have seen that English *may* has the ‘ED’ profile: it can be used to express epistemic flavor, deontic flavor, and no other flavors. *May* thus satisfies connectedness on the orders **E–D–T–C** and **C–E–D–T** (among others) because E and D are adjacent on these orderings. *May* violates connectedness on the orders **E–C–T–D** and **C–E–T–D** (among others) because there is a gap in the meaning space that is covered by the lexical item. Generalizing this to the full data set, the ‘best’ orders should thus be the ones that produce the least number of violations.

The results of the gap analysis are presented in Table 5. In the left column, flavor profiles are listed with their observed frequency in the typological data. (We omit lexifications of a single flavor—E, D, T, C—and of all flavors—EDTC—as none of these produce violations for any order.) 1’s in the body of the table indicate which flavor profiles violate which of the 12 distinct orders. (Each order is equivalent to its mirror image on this metric, so symmetric orders are omitted.) The resulting number of violations in the typological data are listed below each order, with the four best scores in boldface.

<i>attested</i>		edtc	edct	etdc	etcd	ecdt	ectd	detc	dect	dtec	dcet	tedc	tdec	= # of violations
		12	30	19	33	18	14	45	44	33	51	49	30	
ED	2			1	1	1	1			1	1			
ET	1	1	1			1	1		1				1	
EC	7	1	1	1	1			1				1		
DT	15		1		1			1	1		1	1		
DC	2	1					1	1	1	1			1	
TC	8			1		1				1	1	1	1	
EDT	7		1		1	1	1		1		1			
EDC	2	1		1	1		1	1		1				
ETC	0	1	1	1		1						1	1	
DTC	19							1	1	1	1	1	1	

Table 5: The number of observed violations of connectedness for each linear order.

Several important observations emerge from this gap analysis. Most importantly, we observe that there is no order that produces *zero* violations. While some orders are certainly better than others, there is no order that explains the data completely; this is thus an argument in favor of a richer structure than a one-dimensional ordering. This being said, we can nevertheless extract some generalizations about what makes some orders better than others: first, there is a general preference to keep E on an edge, due to the

high frequency of the DTC profile; second, there is a general preference to keep D and T adjacent in the ordering, due to the high frequency of the DT profile. Beyond these generalizations, however, it is notable that there is no good argument for the order of the root flavors relative to E. Specifically, we note that the two best orders (E–D–T–C and E–C–T–D) in fact have the root flavors in mirror image orders. The fact that they have a similar number of violations can be attributed to the similarly high frequency of the EDT profile and the EC profile, respectively.

5.2 Frequency analysis

In our second analysis, we address more directly which flavor profiles appear more or less frequently than one would expect. In particular, we wish to control for the fact that some modal flavors may simply be more common than others. For example, we observed above that the DT profile is frequent, but in principle this could be explained as a statistical epiphenomenon if D and T turn out to be independently frequent.

In Table 6, we thus compare the OBSERVED count of each flavor profile to its EXPECTED count based on the overall frequency that each flavor can be expressed. Expected values are calculated as follows. First, for each of the four target flavors, we count the number modal items that can express that flavor in the observed data. This provides the following counts: E: 107, D: 99, T: 89, C: 83. We divide by the total number of modal items that can express at least one of these flavors—197—to get the probability $P(F)$ that a given modal item expresses a given flavor F . The expected frequency of a given flavor profile is then the product of $P(F)$ for each flavor F expressed by the flavor profile, times the product of $1 - P(F')$ for each flavor F' not expressed, times the total number of items. In Table 6, profiles that appear more frequently than expected are highlighted in green and those that appear less frequently than expected are highlighted in red.

	O	E	O/E
E	58	17	3.4
D	22	14	1.5
T	9	12	0.8
C	15	10	1.5
ED	2	17	0.1
ET	1	14	0.1
EC	7	12	0.6
DT	15	12	1.3
DC	2	10	0.2
TC	8	9	0.9
EDT	7	14	0.5
EDC	2	12	0.2
ETC	0	10	0
DTC	19	9	2.2
EDTC	30	10	2.9

Table 6: Observed compared to expected frequency for each flavor profile

We are able to observe a number of general tendencies in the data. First, we observe that there appears to be a ‘one-or-all’ tendency: on the whole, modal items tend to ei-

ther specify a single flavor or allow wide ambiguity. Among the individual flavors, E is particularly likely to be lexified alone and T is the least likely. Root flavors (DTC) are commonly colexified. Although colexification of just two flavors on the whole is uncommon, the notable exception is DT, as observed above in the gap analysis.

Above, we outlined a number of methodological choices that were made in the treatment of the data. It turns out, however, that these choices have relatively little impact on the generalizations above, suggesting that they are relatively robust. To establish this, the frequency analysis was redone with two alternative sets of data. First, we conducted the analysis with a roughly similar data set, but with different choices of how to count colexifications. Specifically, we conducted the analysis on the April 2023 version of the Guo et al. (2022) database, including 22 languages from Uegaki et al. (2024), plus an additional 18 languages.⁷ As above, target flavors included epistemic, deontic, teleological, and circumstantial (some of the included languages had no modal items listed for any of these flavors). Differently from above, we (a) counted modals in positive and negative environments as separate items and (b) counted modal items as separate when listed with different morphological inflection in the ‘full_form’ column of the database. The results appear in Table 7a. Second, we conducted the analysis with native language judgments from students in an MA-level Semantics 1 class, representing seven native languages.⁸ In a homework assignment on the topic of modality, students were asked to provide the flavor profile of four modals from their native language, and justify each judgment with an example sentence. A few responses were omitted—for example, if the example sentences clearly displayed misunderstanding of the categories—but judgments were otherwise taken at face value. When students with the same native language identified different flavor profiles for the same modal item, the count was distributed proportionally across identified profiles (e.g., if three French speakers identified flavor profile X for a given item and one French speaker identified flavor profile Y, then 0.75 was added to the count of X and 0.25 was added to the count of Y). Results appear in Table 7b, and the full dataset is included as Supplemental Materials.

Notably, these changes to the source of data and the precise choices about counting the data have a relatively small impact in the emerging generalizations. Monolexifications are quite common, especially for epistemic modality. Otherwise, the most frequent flavor profiles are EDTC (all flavors), DTC (all root flavors), and DT (just deontic and teleological).

To further understand the nature of these generalizations, we additionally inspected the flavor profiles that appear in each individual language in the Uegaki et al. (2024) dataset. Specifically, we were interested in whether or not the ‘one-or-all’ tendency reflects a language level parameter—that is, whether some languages have a modal inventory where each item specifies a single flavor, while other languages have a modal inventory where items tend to be ambiguous. Comparing inventories across languages indicates

⁷Languages included: Akan, Basque, Cantonese, Donmari (ISO: rmt), Dutch, Farsi, Gitksan (ISO: git), Goemai (ISO: ank), Greek, Hausa, Hebrew, Hindi, Hinuq (ISO: gin), Hungarian, Hup (ISO: jup), Igbo, Jamul-Tipay (ISO: dih), Japanese, Javanese-Paciran (ISO: jav), Khmer, Kĩtharaka, Korean, Kwaza (ISO: xwa), Lillooet-Salish (ISO: lil), Logoori (ISO: rag), Mandarin, Mani (ISO: buy), Mapudungun, Mian (ISO: mpt), Nuosu (ISO: iiii), Qiang (ISO: qxs), Russian, Tagalog, Thai, Tlingit (ISO: tli), Tundra Nenets (ISO: enh), Turkish, Vaeakau-Taumako (ISO: piv), Vietnamese.

⁸Languages included: French, Italian, Mandarin, Russian, Hungarian, Portuguese, and Georgian.

a.	O	E	O/E	b.	O	E	O/E
E	85	32	2.7	E	5	1.4	3.7
D	48	28	1.7	D	1.5	2.7	0.6
T	21	21	1.0	T	0.25	1.6	0.2
C	32	20	1.6	C	2	0.8	2.5
ED	4	27	0.1	ED	3	3.4	0.9
ET	4	20	0.2	ET	1.83	2	0.9
EC	12	19	0.6	EC	1	1	1
DT	28	18	1.6	DT	6.75	4	1.7
DC	5	17	0.3	DC	1.5	2	0.7
TC	11	13	0.9	TC	0	1.2	0
EDT	5	17	0.3	EDT	2.75	5	0.6
EDC	7	16	0.4	EDC	1	2.6	0.4
ETC	5	12	0.4	ETC	0.5	1.5	0.3
DTC	15	11	1.4	DTC	4.33	3	1.5
EDTC	32	10	3.1	EDTC	5.58	3.8	1.5

Table 7: O/E analysis for (a) different methodological choices with the Guo et al. (2022) database; (b) responses on a Semantics 1 homework assignment.

that this is not the case: rather, inventories tend to be of a similar nature across the 24 languages in the sample. Specifically, all modal inventories had at least one modal item that specified a single flavor, and 22 out of 24 of the modal inventories had at least one modal item that that was three-way or four-way ambiguous between the target flavors. (The exceptions are Japanese and Mapudungun, which nevertheless do have two-way ambiguous items.)

5.3 Flavor change in negative environments

Finally, we returned to the question, raised in Section 3.4, regarding the interaction of modal flavor and the polarity of the environment in which an item appears. In the Uegaki et al. (2024) dataset, 96 modals are described for both positive and negative environments. For these 96 modal items, 33 have the same flavor profile in both uses; 36 lose flavors in the negative use; 12 add flavors in the negative use; and 15 both add and lose flavors in the negative use. If it holds up to further investigation, this is a somewhat striking finding. First, it confirms that the English patterns described in Section 3.4 are not an anomaly: interactions between flavor and polarity turn out to be more common than not. Second, among the flavor changes, it is striking that the most frequent pattern is for modal items to *lose* flavors in negative environments. This being said, it is possible that the apparent flavor changes reflect a methodological artifact. For example, if judgments were collected more systematically for positive than for negative environments, this would result in an apparent loss of flavors, due to the present methodological choice to count ?s as 0s. Nevertheless, this explanation does not seem particularly likely, as Uegaki et al. (2024) state the overt goal of documenting negative as well as positive environments. If these generalizations prove to be sound, it remains an open question why modals frequently change flavor in modal environments, and why this change often involve a loss of flavors.

6 Discussion

Considering a space with four forces and three flavors, the hypothesis of Independence of Force and Flavor (Steinert-Threlkeld et al., 2023) provides a way to narrow down the space from 4095 possible profiles to 105 profiles ($= 15$ flavor profiles $\times 7$ force profiles). Here, we focused specifically on modal flavor, investigating whether there are systematic patterns of colexification among the 15 potential flavor profiles, which would restrict the typological picture yet further. One important take-home from the present work is that there *are* such patterns among these flavors: certain flavor profiles appear more frequently than expected, and some less frequently. Below, we discuss what these patterns tell us about the ontological structure of the space of modal flavor.

6.1 The one-or-all tendency

A first pattern that was observed in the data is a *one-or-all* tendency. Specifically, the most frequent colexification profiles that appear cross-linguistically are ones in which a single flavor is specified, or in which all four target flavors are possible. Together, these five flavor profiles (one third of the total 15), account for more than two thirds of the data (134 out of 197 items).

There are at least two ways to understand this generalization. First, it is possible that at least some part of this generalization may arise from principles of efficient communication. In particular, monolexification and total colexification instantiate the two extremes on a scale, so may arise frequently in inventories that are optimal solutions to a particular trade-off between informativeness and simplicity.

Alternatively, the appearance of a one-or-all tendency may be spurious, the result of ontological constraints on colexification that only affect other flavor profiles. We note in particular that the one-or-all tendency—if it exists—is by no means absolute; in particular, the DTC profile and DT profile also appear across the data with a high frequency. The high frequency of DTC and DT notably motivates the existence of constraints relating to the structure of the meaning space, since, without such constraints, these profiles are symmetrical to all other profiles with three or two flavors, respectively. Crucially, though, constraints on colexification that emerge from the structure of the meaning space only have an effect on flavor profiles that colexify several but not all flavors. In particular, a set containing a single item and the set containing all items are, trivially, also valid natural classes for any meaning space. (This is, for instance, why these five flavors did not appear in the gap analysis, as none of them lead to violations of connectedness for any structure.) The fact that ontological constraints only have a meaningful inhibitory effect on profiles in which several but not all flavors are colexified may in turn create the appearance of a one-or-all tendency.

These two factors are of course not in opposition; it is possible that the existence of a one-or-all tendency reflects pressures related to efficiency, but also a reduced set of several-but-not-all profiles due to constraints related to the structure of the meaning space.

6.2 The epistemic-root opposition

The colexification patterns presented above support the existence of a natural class of root modals, witnessed in the frequent appearance of the DTC flavor profile. Strikingly, this finding provides independent motivation for the epistemic-root opposition, traditionally motivated based on empirical data of an entirely different kind: scope and word-order.

This finding has several theoretical consequences. First, it provides new support for the existence of the root category. Recall, for example, that Portner (2009) claims that the root category is not a natural class (despite having a mechanism to derive it as a primitive in his system). The data from colexification suggest that it is.

Second, the fact that both scopal facts and colexification facts identify the same natural class calls for an explanatory mechanism that derives both observations. Of note, though, patterns of scope and patterns of colexification are often seen as arising from mechanisms of a fundamentally different nature. Patterns involving scope are generally taken to reflect structural properties of a synchronic grammar; in contrast, colexification is often taken to emerge from factors influencing diachronic change (Bybee, 2003; van der Auwera and Plungian, 1998).

As we saw in Section 3.1, a relatively standard explanation of the scopal facts is that different modal flavors are associated with different positions on a syntactic spine. Extending such an analysis, one could in principle imagine an analysis in which such structural representations also underlie patterns of colexification. In particular, in other domains, some cases of colexification (namely, *syncretisms*) have been explained as arising from syntactic containment (Bobaljik, 2012). On theories like Distributed Morphology (Halle and Marantz, 1993) or Nanosyntax (Starke, 2009), a lexical insertion rule may identify a single lexical item as the phonological exponent of a complex morphosyntactic structure; the same item may then also spell out any subpart (Distributed Morphology) or superpart (Nanosyntax) of this morphosyntactic structure that does not have its own dedicated lexical insertion rule. For the case at hand, one could thus posit that different modals spell out different subparts of the syntactic spine.

In light of the colexification data presented here, however, such an analysis proves to be intractable on several fronts. First, the gap analysis in Section 5.1 showed that a linear ordering of modal flavors does not appear to be a viable ontological structure. There is no linear order that produces zero (or close to zero) violations, and, even more strikingly, there is no strong evidence for an ordering of the root flavors relative to epistemic flavor: the two best structures have root flavor in mirror image orders. Relatedly, a cartographic analysis has no way to explain the relative infrequency of 2-flavor colexifications. On such an analysis, pairs of adjacent flavors would correspond to minimal chunks of syntactic structure, so 2-flavor colexifications would be wrongly predicted to be at least as common as ‘mega’-colexifications, of all flavors or all root flavors. As we have seen, this is not the case.

Second, a more general problem is that morphological theories are designed to describe phenomena where different morphological forms cleanly partition a space of semantic functions: while the same form may be used to express two different meanings, each meaning is associated with a only a single form. This property emerges from the fundamental design of these morphological theories—specifically, from the Elsewhere Condition, on which lexical insertion is deterministically resolved via application of the most specific rule. For inventories of modal items, this simply does not seem to be an

appropriate characterization of the data. In English, for example, the colexifications observed in (5) are highly overlapping, and a similar situation holds across the languages in our data set. A morphological explanation in terms of syntactic containment and the Elsewhere Principle is thus not feasible for the domain of modal flavor colexification.

If the colexification facts do not have their origins in an underlying hierarchical representation, then the converging patterns of scope and colexification suggest that the scope facts might not either. An alternative explanation is that both the colexification and scopal patterns have their origins in semantic features that characterize epistemic and root modality as natural classes. On a standard Kratzerian view, this feature could simply be the identification of the modal base as either epistemic or circumstantial. On this view, colexification can be modeled with a presupposition carried by the item in question, as described in Section 3; modal items that colexify sets of root flavors would be posited to carry a presupposition of a circumstantial modal base. The scopal facts, for their part, may have pragmatic origins, arising from the conceptual difficulty or the limited utility of communicating the meaning corresponding to the other scopal configuration. For example, we observed in Section 3.1 that the sentence in (25) only appears to allow a reading in which tense takes scope over the modal verb: the sentence asserts that circumstances *at the time of the event* obliged Mary to take the train. But, the alternative scopal ordering is close to nonsensical, entailing that circumstances at the time of utterance somehow oblige an event to have already happened. For this particular example, no further syntactic explanation is necessary.

(25) Mary had to take the train to go to Paris.

Past > □

The literature on modality has of course considered such explanations, with some authors concluding that conceptual or pragmatic factors alone are not sufficient to explain the data. For example, Hacquard (2010) shows that there are cases in which a simple modal verb is unable to express a particular scopal ordering despite the existence of paraphrastic strategies that have no difficulty in expressing the intended conceptual meaning.

In light of such arguments, a possible compromise position is that conceptual and pragmatic factors are the diachronic *origin* of these patterns, but that the scopal effects have become part of the grammar over the course of language evolution, thus eventually arriving at a synchronic representation that encodes these preferences as genuine structural constraints. We note that such an explanation is of a somewhat different nature than the hierarchy proposed on traditional cartographic approaches. For Cinque (1999), the proposed syntactic hierarchy is itself an element of Universal Grammar. On the compromise position outlined here, human language has a tendency to posit hierarchical syntactic structures, but it is an underlying pragmatic bias that provides the scaffolding on which these structures are built. Such a position might therefore predict that while epistemic modals are systematically higher than root modals cross-linguistically, hierarchical distinctions among root flavors—with less clear pragmatic motivations—may still exist but would be subject to greater cross-linguistic variation.

6.3 Subclasses of root modals

A second notable natural class that clearly emerged in the colexification data is the grouping of deontic and teleological modality. Notably, as described in Section 3.2, the association of deontic and teleological modality is predicted on several typologies of modal

flavor. First, the DT profile corresponds to the subset of *priority* modals, characterized by an ordering source that allows one to rank some worlds as better than others (Portner, 2007, 2009). Second, it corresponds to the subset of *participant-external* modals, relating to circumstances external to the participant in question (van der Auwera and Plungian, 1998). Like for the case of root modality, the colexification data provide converging motivation for a semantic class previously proposed based on entirely different evidence (for the case of priority modals, parallels with imperatives).

As discussed in Section 3.2, the natural classes of priority modals and participant-external modals do not necessarily coincide when we expand the set of modal flavors beyond the four studied here. In particular, bouletic modality should be classified as a priority modal on Portner’s analysis (thus patterning with D and T), but as a participant-*internal* modal on van der Auwera and Plungian’s analysis (thus patterning with C). One may in principle be able to evaluate these competing proposals by assessing the frequency that bouletic modality is colexified with the other flavors. As it turns out, though, the colexification data do not support either of these analyses over the other. Of the 43 items that are able to express bouletic modality in the Uegaki et al. database, only one has the DTB flavor, and zero have the CB flavor. The colexification of bouletic flavor thus does not provide evidence for either Portner (2007) or van der Auwera and Plungian (1998), although, this may be due to sparseness of data. Of the 47 items that can express bouletic flavor, the four profiles that appear more frequently than expected are: total lexification EDTCB (14 items), monolexification B (12 items), total lexification of root flavors DTCB (6 items), and the profile TCB (4 items). All other profiles appeared for only 0, 1 or 2 items. The colexification data involving bouletic flavor thus appear to reflect the larger one-or-all tendency, and also confirm the classification of B as a root flavor, but they don’t suggest any other clear further natural classes.

6.4 Evaluating different ontological structures

Section 3 presented several ontological structures that could hold of the underlying representation of modal flavors. Here we evaluate these possible structures in light of our colexification data.

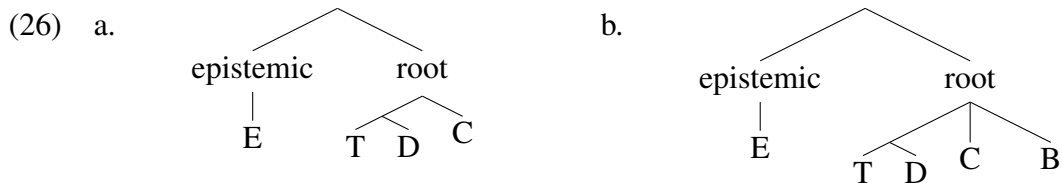
First, reprising the argumentation in Section 6.2, we observe that a *linear, one-dimensional ordering* is not viable. As we saw in Section 5.1, there is no linear order that produced zero violations on the gap analysis; moreover, there is no single order of the root flavors relative to epistemic flavor that is clearly better than others. A linear ordering is also unable to explain the relative scarcity of 2-flavor colexifications, which should in principle be possible for any pairs of adjacent flavors in the linear ordering.

Second, *semantic maps* also turn out to be ill-equipped for modeling modal flavor colexifications. As discussed in Section 3.3, semantic maps are defined in terms of pairwise relations between meanings; it is thus impossible to define a natural class containing three or more elements without positing subclasses containing pairs of elements. But this is exactly the situation that we find with modal colexification: apart from the DT flavor profile, 2-flavor colexifications are very low. The challenge for semantic maps is perhaps best illustrated with epistemic flavor. On the one hand, epistemic flavor is clearly not *disconnected* from other flavors: indeed, the second most frequent flavor profile is EDTC, which colexifies epistemic modality with all other flavors. On the other hand, 2-flavor

colexifications with E are extremely low across the board: it is thus impossible to identify what epistemic modality is connected to. Modal flavor colexifications thus cannot be modeled as a graph of pairwise relations.

A featural analysis, on the other hand, is able to capture the colexification data in a motivated manner. The mere fact that some featural analysis can found to model the colexification data is in fact trivial, as one can simply posit the existence of a feature for any subclass of flavors that appears at a rate higher than expected. What gives the featural analysis explanatory weight in the present case is the observation that the features that emerge from the colexification data ‘make sense,’ coinciding with features independently motivated in the literature. For the four flavors included in the main analysis here, two primary features appear to be in play: first, a feature relating to the modal base, separating epistemic modality from root flavors; second, a feature that identifies a property of the ordering source which yields a natural class of deontic and teleological modality.

In the current case, these features appear to be nested; they can thus be represented graphically in a tree structure, as shown in (26). The tree in (26a) displays only the four flavors included in the two main analyses here; the tree in (26b) also includes bouletic flavor based on the somewhat sparse data discussed in Section 6.3. Because the current data is limited to only four flavors (and only one subflavor of epistemic modality), further questions about the nature of this structure necessarily remain open. For example: as described in Section 3.3, when features are organized in a feature geometry, then some features may be dependent on other features (i.e., an element can only bear feature Y if it also bears feature X). In the present case, it remains open whether or not the feature that forms a natural class of T and D is dependent on a circumstantial modal base, or whether it is also relevant to the categorization of subclasses of epistemic modality. This cannot be addressed by the information in the present data set, which only identifies a single kind of epistemic flavor, so presents nothing to categorize.



6.5 The limitations of the data

The nature of the data set means that there are a number of further questions that necessarily remain open. For example, as described in Section 4.2, different contexts that ostensibly exemplify the same force-flavor combination are sometimes judged by the same speaker to be compatible with different sets of modal items. This is compatible with the possibility that there are further distinctions in the lexical semantics of modal items that are not captured by the assumed categorization into three modal forces and five modal flavors. However, the nature of these further distinctions, and the degree of their cross-linguistic variability remains open.

For example, as described in Section 3.2, it has been proposed that epistemic flavor can be further subcategorized as subjective or objective (Nuyts, 2001). Elsewhere, Phillips and Kratzer (2024) have proposed that epistemic modality can be categorized regarding whether it is grounded in an actual or an epistemic anchor. It is possible that distinctions

such as these may be responsible for some of the variation across contexts observed in our replication data. However, just as the phonology of two languages may vary in the number of categorical distinctions that are made in phonological space, it is perfectly feasible that the semantics of two languages may vary in the number of categorical distinctions that are made in semantic space.

Moreover, when discrete categories are imposed a continuous underlying space, different language may in principle vary not only with respect to the number of categories, but also with respect to where the boundary between different categories lies. Such is known to be the case in phonology: for example, voice onset time is a continuous phonetic variable that distinguishes the voicedness of stops in onset position; languages vary not only in the number of voicing categories, but also in the dividing line between these categories (Lisker and Abramson, 1964). By analogy, it is possible that at least some of the features that distinguish different flavors of modality are built on a continuous semantic space. The literature on modality and related domains suggests a number of contenders for such continuous semantic variables. For example, in an adjacent domain, Oomen and Roelofsen (2023) argue that strength of speaker belief and strength of contextual evidence are two (continuous) dimensions that influence the availability of different polar question forms cross-linguistically. Similarly, Phillips and Kratzer (2024) suggest that modal semantics is sensitive to the ‘normality’ of a domain restriction, a concept that can likely be spelled out in terms of a continuous variable. Pending a cross-linguistic study that deliberately manipulates such continuous variables, it remains open whether they are indeed incorporated into the lexical semantics of modal items, and how much cross-linguistic variation exists regarding the boundary between discrete categories.

These are questions that cannot be tested with the present data set, which simply presupposes a set of standard flavors commonly proposed in the literature: epistemic, deontic, teleological, circumstantial, and bouletic.

7 Conclusion

Modal items are known to often express ambiguity with respect to flavor specification. Here, we took advantage of the recently created database of Uegaki et al. (2024) to uncover the natural classes that emerge from modal flavor colexification, and to evaluate the ontological structures that are compatible with these natural classes.

In a first step, we replicated the Uegaki et al. database for three of their 24 languages, in order to know how to treat the large number of undetermined entries that appear across the database as a whole. For Hebrew, Mandarin, and Russian, we thus compared the entries in the Uegaki et al. database, which consists largely (though not exclusively) of positive judgments coming from a translation task, to the results of a methodology in which judgments were exhaustively collected for each item-meaning pair. The results of the replication provide evidence that a lack of direct evidence of acceptability of an item-meaning pair in the Uegaki et al. database can generally be taken as indirect evidence of its unacceptability. This finding might apply more generally to other studies that adopt translation-in-context methodologies, although this is likely to depend on the specific details of the elicitation procedure.

We then conducted two quantitative analyses regarding the modal flavors that appear in the database. First, we asked which linear orderings produce the smallest number of

violations of connectedness. Second, we compared the frequency of each flavor profile to its expected frequency. We showed that the generalizations that emerge are relatively robust, appearing in two replications of the study with different methodological choices.

Several notable patterns of flavor colexification appeared cross-linguistically. First, we observed a one-or-all tendency: modal items tend to either identify a single modal flavor, or be highly ambiguous. We observed that this fact could emerge either from pressures on efficient communication or on feature sets that rule out most two-flavor combinations. Second, we observed that an epistemic-root divide appears in colexification patterns, thus mirroring a distinction that is traditionally motivated based on entirely different evidence. We discussed the possible mechanisms that could explain both the colexification facts and the scopal facts that traditionally ground the epistemic-root distinction; we proposed a compromise position on which semantic and pragmatic factors feed syntactic constraints. Finally, we observed a high affiliation between deontic and teleological flavors; we discussed two recent proposals that form a natural class of these two flavors, with the conclusion that neither one manages to perfectly identify the observed natural classes when bouletic flavor is taken into account (although the data is sparse).

Finally, we evaluated these natural classes with respect to different ontological structures. Notably, we observed that neither one-dimensional orderings nor semantic maps have sufficient power to capture the descriptive generalizations of flavor colexification, despite corresponding to structures that have been proposed elsewhere to analyze modal flavor. These results argue in favor of an analysis in terms of semantic features, although the precise nature of these features and their relation to each other remains open.

A number of further questions also remain open. First, we observed in Section 5.3 that the modal flavors expressible by a particular item appear to often change in negative environments, with a particular propensity for modal items to lose modal flavors under negation. If this generalization stands up to further scrutiny, it remains an open question why modal flavor is sensitive to the polarity of the embedding environment. Second, because our data set assumes a small set of primitive modal flavors, it remains an open question whether languages vary in the number of flavor categories they encode and the position of these categories within semantic space.

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